7 There are only red counters, green counters and blue counters in each of two bags.

In bag **A**,

number of red counters : number of green counters : number of blue counters = 10 : 9 : 6

In bag **B** there are

20% more red counters than in bag **A**
100% more green counters than in bag **A**
$\frac{1}{3}$ more blue counters than in bag **A**

Which bag has the greater fraction of green counters, bag **A** or bag **B**?
You must show how you get your answer.

Use proportional counts 10 red, 9 green, 6 blue.

Bag B red: 10 × 1.2 = 12

1 mark
Increase red by 20%.

Bag B green: 9 × 2 = 18

Bag B blue: 6 + 6 ÷ 3 = 8

1 mark
Double green; add one third to blue.

Bag A: $\frac{9}{25}$ Bag B: $\frac{18}{38} = \frac{9}{19}$

Same numerator: 9/19 is greater than 9/25.

1 mark
Compare fractions of the TOTAL in each bag, not counts.

Bag B has the greater fraction of green counters.

1 mark
For any common scale factor, the same fractions result.

(Total for Question 7 is 4 marks)

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